Experiment 17: Docker Volumes and Port Mapping

Objective

To understand Docker **data persistence using volumes** and **communication using port mapping**, ensuring container data is not lost and services are accessible externally.

Introduction

Docker containers are **ephemeral**, meaning data inside a container is lost when the container is removed. Docker **volumes** provide persistent storage by saving data outside the container. **Port mapping** allows external systems (host machine or network) to access services running inside containers.

Part A: Docker Port Mapping

Step 1: Run Nginx container without port mapping

```
docker run -d nginx
```

This container runs, but cannot be accessed from the host browser.

Step 2: Run Nginx container with port mapping

```
docker run -d -p 8080:80 nginx
```

Explanation:

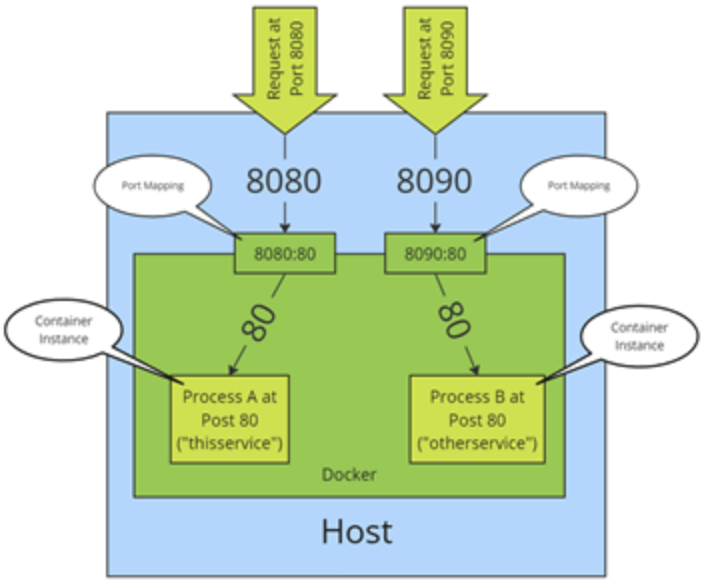
- `-p 8080:80`
  - 8080 → Host port
  - 80 → Container port

Step 3: Verify

Open browser:

```
http://localhost:8080
```

You will see the Nginx welcome page.



Part B: Docker Volumes (Data Persistence)

Step 1: Create a Docker Volume

```
docker volume create myvolume
```

Verify:

```
docker volume ls
```

Step 2: Run container using volume

```
docker run -it -v myvolume:/data ubuntu bash
```

Explanation:

- `-v myvolume:/data`
  - myvolume → Docker volume
  - /data → Container directory

Step 3: Create file inside container

Inside container:

```
cd /data
echo "Docker Volume Persistence Test" > test.txt
exit
```

Step 4: Run new container using same volume

```
docker run -it -v myvolume:/data ubuntu bash
```

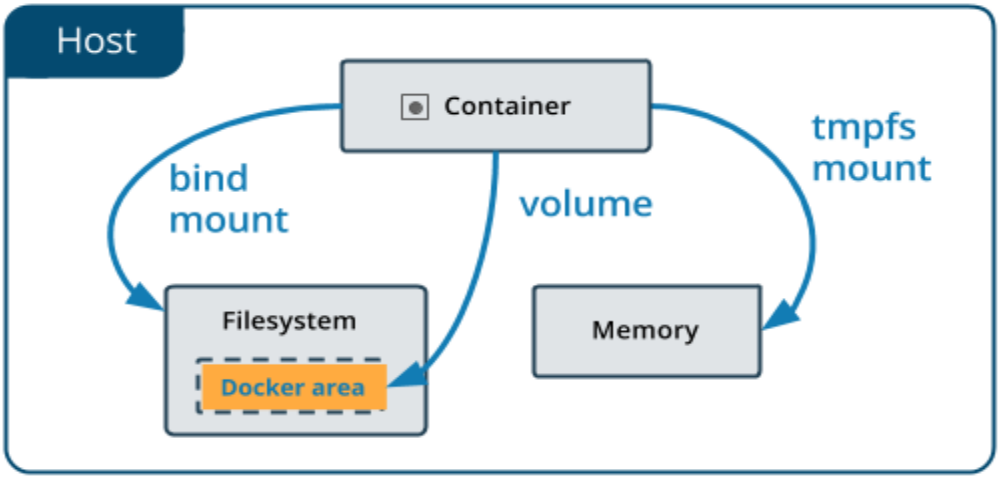
Check file:

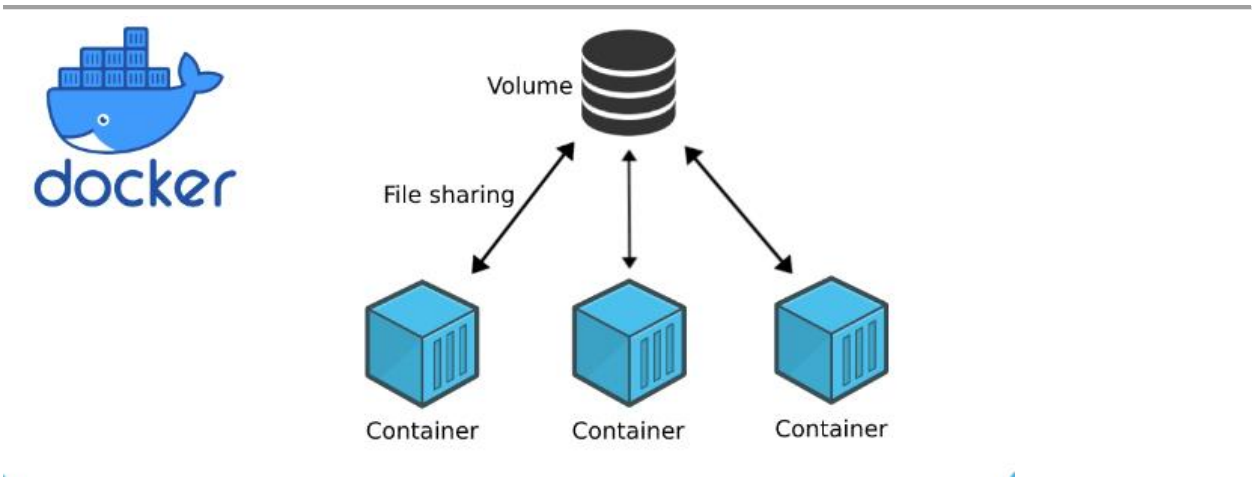
```
cat /data/test.txt
```

Output:

```
Docker Volume Persistence Test
```

This proves data persists even after container removal.





Part C: Combined Example (Port Mapping + Volume)

Run container with both volume and port mapping:

```
docker run -d -p 8081:80 -v myvolume:/usr/share/nginx/html nginx
```

Add HTML file:

```
docker run -it -v myvolume:/data ubuntu bash
echo "<h1>Persistent Web Page</h1>" > /data/index.html
exit
```

Open browser:

```
http://localhost:8081
```

Persistent webpage will appear.

Verification Commands

Check running containers:

```
docker ps
```

Check volumes:

```
docker volume ls
```

Inspect volume:

```
docker volume inspect myvolume
```